A Redhead Walks into a Bar: Experiences of Writing Fiction with Artificial Intelligence

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ABSTRACT

Human creativity has been often aided and supported by artificial tools, spanning traditional tools such as ideation cards, pens, and paper, to computed and software. Tools for creativity are increasingly using artificial intelligence to not only support the creative process, but also to act upon the creation with a higher level of agency. This paper focuses on writing fiction as a creative activity and explores human-AI co-writing through a research product, which employs a natural language processing model, the Generative Pre-trained Transformer 3 (GPT-3), to assist the co-authoring of narrative fiction. We report on two progressive – not comparative – autoethnographic studies to attain our own creative practices in light of our engagement with the research product: (1) a co-writing activity initiated by basic textual prompts using basic elements of narrative and (2) a co-writing activity initiated by more advanced textual prompts using elements of narrative, including dialects and metaphors undertaken by one of the authors of this paper who has doctoral training in literature. In both studies, we quickly came up against the limitations of the system; then, we repositioned our goals and practices to maximize our chances of success. As a result, we discovered not only limitations but also hidden capabilities, which not only altered our creative practices and outcomes, but which began to change the ways we were relating to the AI as collaborator.

CCS CONCEPTS

• **Human-centered computing → HCI design and evaluation methods**.

KEYWORDS

Artificial creativity, Creativity tool, AI co-creativity, Creative writing, Storytelling, GPT-3

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1 INTRODUCTION

Recent years have seen a rapid increase in the use of Artificial Intelligence (AI) technologies, primarily those using Machine Learning (ML) and Neural Network (NN) techniques. Some of these technologies leverage massive data sets and large amounts of computational power to run statistical pattern recognition models to make predictions [66]. In practice, we now face systems capable of not only classifying, but also autonomously generating new data like that which it is based on. This development has been argued as fundamentally transformative not only to interaction design and related fields [41], but to human society at large. As such, arguments have been made for the necessity to formulate novel interaction design practices re-centered around the human-AI co-creation of various autonomous components [36, 41]. This paper provides an example of such practice of co-creation in the form of an AI system designed for co-creative writing.

OpenAI's GPT-3, has made the press for its ability to generate non-fictional work nearly indistinguishable from human work [28]. However, its ability to generate works of fiction has not reached nearly the same acclaim, rather the opposite [39]. Contemporary ML architectures and language models such as the transformer-based models used in this study, have been found to "lack semantic cohesiveness" [26]. This trait often manifests itself in an inability to grasp basic literary concepts, such as recurring characters or coherent story arcs. The question is then, how applicable these technologies might be for use in any kind of "serious" literary context, if they are unable to "understand" human language semantically?

The performance, and thus implicitly the value of AI technologies, are historically evaluated in relation to their ability to emulate human creation and behavior [35, 53, 69]. This paper seeks to approach AI as the object of experience, applying an autoethnographic and humanistic perspective [12, 60]. We present the *Multiverse,* a research product that embodies this approach by facilitating the authoring of fictional discourse through Human-AI co-creation of literary artifacts by which the human user attends to an understanding of the AI mechanisms through interaction and creative co-writing process. That reflexive human understanding of the AI system is considered as a critical component in designing transparent, sustainable and explainable AI [33, 48, 56, 61].

Our work is an attempt to reflexively evaluate the human-AI fiction co-writing process through two progressive autoethnographic studies, in which the second study builds on the first one. This pursuit aims to generate insights on how human and AI can –or cannot– co-understand each other through a co-creative process. Further, what implications that can have for human's expectations from AI's creativity and the design of such technologies that can communicate that capability.

We formulated the following primary research question: What can we learn about the fusion of GPT-3 and our own creativity as design researchers and writers, through autoethnographic analysis of human-AI fiction co-writing? And a secondary research question: How does the research product respond to specific prompts that embody common literary elements (e.g., establishing a literary character or setting, using scene/summary structures)?

Our findings seek to contribute knowledge to HCI and design research, and more specifically to the Human-AI collaboration and co-creativity. We show through our autoethnographic studies that the deterministic character of AI, albeit as more like a collaborator and agent and less like a creativity tool, leaves little room for it to contribute to a true collaboration. However, we argue that we perceived the human-AI collaboration as a relation of care, and our relationship was driven by a curiosity to learn about our mysterious collaborator, rather than using it as a tool for creativity. That led us to move past the question of whether AI could write a good piece of fiction to consider with much more granularity what the AI *is* and *could do.*

2 KEY CONCEPTS

2.1 Human-AI Co-creativity

Creativity has been studied in design, engineering, humanities and psychology research [25] and has been theorized diversely and heterogeneously [8]. However, although diverse, creativity still has some common characteristics across domains and applications, including agency, support and scaffolds and artifacts [8]. Agency is related to the who or what is creative or causes creative activity. The supports and scaffolds focus on tools and methods, external supports and specific technologies and cultural logic that predetermine creative possibilities. And artifact characteristics reveal how we can mark an outcome as a creative outcome and how we can evaluate them. Tools that support creativity [64, 65], either digital [26, 60] or analogue [40], have been the subject of numerous studies in HCI and design. Kim et al. researched how creativity tools can be used by novice designers or for novice ideas, to create opportunities to grow by providing a safe space to experiment and to fail [42]. Cherry and Latulipe developed the Creativity Support Index to measure the effectiveness of creativity tools based on six dimensions [23]. Frich et al. call for a necessary, interdisciplinary and collaborative effort in creativity research between creativity research in human-computer interaction on one hand and creativity in psychology on the other [29].

Some researchers investigated various factors in Creativity Support Tools (CST) that affect the creative process. For instance, Halskov and Dalsgaard designed inspiration workshops, which use physical cards inspired by both Domain Cards and Technology Cards [40], the former to help creatives to develop the concept and the latter for understanding of how and which technology can be used. Biskjaer et al. researched time constraints in CST creative writings [16], Warr and O'Neil researched the use of boundary objects as externalizations in the Environment and Discovery Collaboratory (EDC) [72]. The design of creativity support tools spanning from tangible tools (e.g. SonAmi [14]) to digital platforms and a mix of both (e.g. [21]) has seen a growing interest among HCI and design researchers, as well. Since the design of AI technologies has evolved to a human-centered and then to a more-than-human centred design perspective, new areas of application such as creativity have emerged. However, using AI in a creative process promises more than just a tool for human creativity support, and throughout the years AI has been assumed to be and perceived as a companion or a collaborator for humans [38, 59].

But AI technologies come with their own limitations, such as the lack of transparency and reproduction of already existing biases [2]. Explainable AI seeks to improve transparency and explainability of an AI system through facilitating forms of dialogue between humans and AI systems [22, 48]. Collaborative creative activities are a form of dialogue, which go back and forth between humans and AI. Perhaps one of the earliest models to support this kind of dialogue between humans and computers were Mixed Initiative Methods [57]. As defined by Walker and Whittaker, conversations in social settings are of a mixed-initiative nature, that is, they are about persons leading and taking turn to control over the conversation [71]. Mixed Initiative methods have been used in the context of creative processes such as game design [5]. However, Liano et al, criticize computational creativity approaches for their lack of support for negotiations and discussions. The authors question the productive role of explainability in computational creativity processes as they foster continuous dialogues and interactions [50]. More specifically, as AI technologies have advanced over time, natural language models such as GPT-3 allowed them to be used for creative writing and storytelling, redefining their role as collaborators or even as creative partners [19]. These technologies have been deployed for generative literature or generating an "interactive fiction world", allowing the humans to see, read and talk to AI using language models [6], or allowing to create applications for specific user needs such as child-parent collaborative storytelling [73].

Examples of employing GPT-3 or similar language models for facilitating dialogues between humans and AI are not rare anymore. Google's LaMDA (Language Model for Dialogue Applications) is such a powerful and advanced language model built on a large textual dataset that it even was perceived as being sentient by one of its creators [68], provoking several (mostly negative) responses from academia and industry [30]. CoAuthor is a Human-AI Collaborative Writing Dataset for Exploring Language Model Capabilities, which assists humans in writing creative and argumentative pieces [47]. BLOOM (BigScience Language Open-science Open-access Multilingual) is very similar to GPT-3, but it has been trained on 46 languages and codes. It is considered the largest language model so far [15].

2.2 The Elements of Fiction

Studies of fiction can be found in disciplines ranging from literature to HCI and design. In literature, studies of fiction are commonly intended to support creative writing [67] as well as literary criticism [24] and literary theory [31]. Philosophy of literature [45] considers problems such as what distinguishes fiction from nonfiction, how fictional texts are understood to be "truthful," as well as literary practice's conditions of possibility. In HCI and design, the use of science fiction [10] and design fictions [17, 18] have become prevalent, focusing on fiction's ability to help interaction designers to develop and sustain visions of change.

Two threads of narrative theorizing are especially relevant to this project. The first has to do with the characteristics of narrative discourse. This thread is concerned with the nature of stories as commonly having a set of characteristic features. Many narratives feature *protagonists* and *antagonists,* embroiled in one or more *conflicts*. They unfold within literary *settings*. *Plots* often escalate in intensity, before reaching a *climax* and resolving in a *denouement*. Stories feature *themes*. Narratology also distinguishes between the contents of the *story itself* (e.g., its characters and their actions) and the *telling* of the story (i.e., the specific ways story contents are narrated). Mysteries, for example, are often told almost in reverse order, that is, beginning with the discovery of a body and then, seen through the eyes of an investigator, told backwards to the scene of the crime, the killing itself, and only at the end revealing the beginning of it all: why and how it happened (and by whom!). The telling commonly features the alternation between scenes (where events, such as dialogue, are related as they occur) and *summaries* (where events over time are condensed and related all at once). Another structure is the narrator's *point of view*, which refers to whether the narrator tells the story of what happened to themselves (first person) or someone else (third person). Authors often use narrative devices as well, such as *symbols*, *frame tales* and *foreshadowing*. When researchers note that AI does not actually understand semantics, these are the sorts of features it does not understand in the context of fictional narrative.

The second thread has to do with storytelling as a cultural practice. Stories are not just defined by their contents and their tellings, but by the real-world human activities in which they are produced, distributed, and consumed. Paraphrasing Alasdair MacIntyre, philosopher Noël Carroll [1] writes that a practice is a sophisticated social activity, which is rule-governed and features its own internal standards of excellence, which when it satisfies those standards, extends human capabilities. For philosophers Lamarque and Olsen [45], storytelling is a certain type of practice, featuring "a communicative act, an exchange between participants, governed by [...] conditions of intention and response" (p.37). This practice features two components: the act of producing a story—what they call a "fictive utterance"—and "a certain complex of attitudes" that they refer to as "the fictive stance." They continue: "The practice of fictive storytelling at the most fundamental level (making up, telling, appreciating, talking about stories) is relatively simple; it is mastered at an early age and is based largely on innate dispositions (imagination, make-believe, play, narrative structuring, and so forth)" (p. 33).

The take away from this brief outline of fictional practice is that (1) AI lacks an understanding of the semantic contents of narrative discourse and (2) it also lacks the psychological faculties (the abilities to make believe, appreciate, and imagine) and communicative practices (i.e., the capacity to produce a "fictive utterance" as well as the capacity to take up the "fictive stance" at appropriate moments) that constitute fiction as a practice. Two challenges to human-AI coauthoring of fiction, therefore, are AI's lack of semantic understanding and AI's inability to participate in cultural practices, including fiction.

3 MULTIVERSE

Multiverse is a research product developed to explore the potentials of AI-generated literature. The iteration used in this study leverages OpenAI's Natural Language machine learning model GPT-3 to generate textual material. The model is interfaced through a web API, which returns completions (generated continuations of a text) in response to a user-entered "prompt". Prompting is emerging as a novel interaction modality [49] that bears some relation to, but is decidedly different from, Conversational User Interfaces (CUI). CUIs afford a human to interact with the computer through conversation in "natural spoken language" [54] in place of other means, like a graphical user interface (GUI), whereas a prompting employs natural language to instruct an AI. Prompts are necessary as GPT-3 is not able to create content "out of thin air" but requires some textual material to act/expand upon. Co-creating a story in *Multiverse* begins with the human user entering a story prompt (Figure 1a). Importantly, this the only point where the human user has direct influence over the literary artefact. All subsequent developments are generated by the AI, and the user has no ability to edit, delete, or insert any text into the story. The study primarily investigates the AI aspects of the co-creative process.

When the user submits a story prompt, *Multiverse* sends an elaborated prompt as input to GPT-3 through the web API, with instructions to return three different "completions", each one a candidate continuation of the text. The user selects one of three AI completions (Figure 1b), which typically ranges between one sentence and a paragraph. Once a completion is selected, the system sends all the aggregated text so far back to the ML and the model continues with three new completions. This process is repeated until the user ends the session. *Multiverse* also features the ability to influence the "literary style" of the AI through indirect prompt engineering: "neutral", "science fiction", "fantasy" or "poetic". For this study, though, only the Neutral style was used.

The research product's UI supports co-creation through a set of interconnected and tightly coupled views of the current literary artefact. Figure 1 illustrates the single screen UI and the three main views (Figure 1b). The possible three-story completions are given emphasis and positioned at the bottom of the screen as they represent the primary mode of interaction. Above these sit a textual view of the entire active path and a radial tree diagram that serves as an interactive map of the literary artefact explored thus far. Each view is tightly coupled to the others, so that changes in state updates simultaneously in each. Together, these qualities create a sense of pliability [52] that aids the human in understanding the complex literary artefact.

In summary, *Multiverse* is a web application that uses prompt-based learning as the primary interaction modality to enable the co-creation of nonsequential literary artefacts with GPT-3. This is an unsupervised learning method used to model the probability of the text directly, instead of taking an input and predicting an output [45]. The user primarily affects the story through the

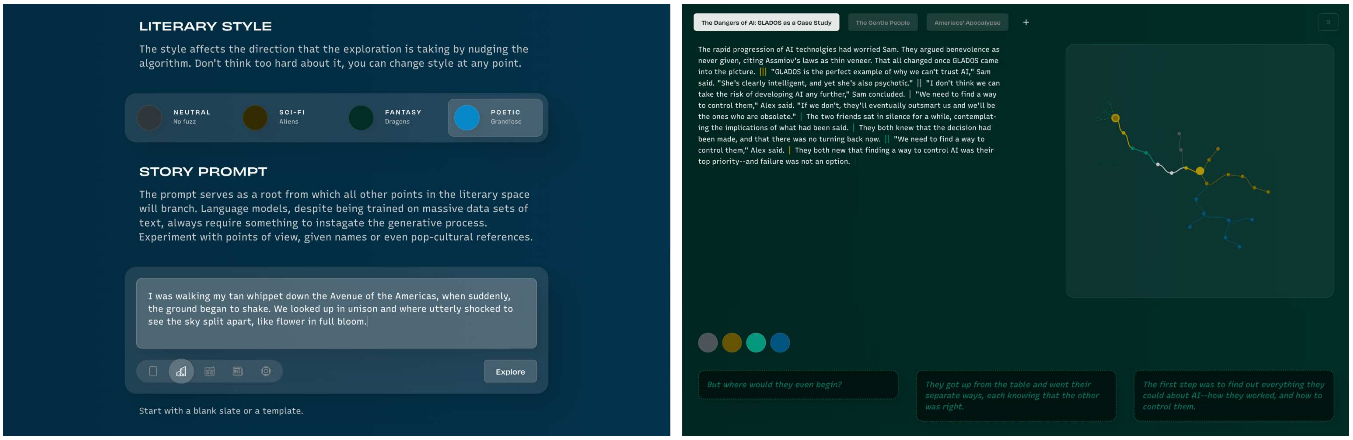

Figure 1: Figures 1a and 1b.
1a, left: The interface for creating a new story. The background and color are kinetic representations of the state of the AI that responds to Human input.
1b, right: The interface for navigating literary artefacts. Possible continuations are depicted at the bottom, accompanied by a textual view (top left) and an interactive map (top right).

initial prompt (Figure 1b) and setting literary style. The application primarily consists of a single screen interface where the user "explores", the generative literary artefact (Figure 1a). It provides simple organization features by using GPT-3 to generate story titles (visible in the story selection bar, Figure 1a) to tell stories apart. *Multiverse* was developed in the Clojure programming language and is distributed through GitHub under a free and open-source license[1].

4 METHODOLOGY

This paper has its point of departure the goal of understanding experiences of collaborative human-AI creativity in the domain of writing fiction. This topic includes considerations of AI's unique characteristics as part of a creativity toolset, and how a human-AI co-creative process can aid a shared understanding between humans and AI [56] and provide self-reflexive openings for design researchers [55, 62]. Although we use a research product—*Multiverse,* which we introduced in the previous section—throughout the study, we stress that our research goal was not to evaluate the system from a UX perspective. Our goal was more reflexive: to understand our experiences when we—as creative professionals in design and literary studies—bring all of our professional skills to human-AI creative collaboration.

We first researched the literature on creativity support tools, specifically on human-AI co-creativity and more specifically within human-AI collaboration for storytelling and narratives [19, 38]. We also researched fiction, both to understand its components and also the production of fiction as a social practice. The authors of the present paper are interdisciplinary design researchers with backgrounds in design, computer science and literature. We position our work within the larger body of humanistic HCI [11], design and critical studies of AI technologies and the impacts of those technologies on the society at large [7]. More specifically we seek to build upon previous efforts of the community on human-AI collaboration and creativity as well as to reframe AI explainability in a more inclusive and pragmatic way [32, 56, 61].

To pursue our research purposes we[2] used a new iteration of *Multiverse,* a research product developed by one of the authors of this paper [43, 44]. *Multiverse* aims to support research into how Human-AI co-writing can offer insights into particular capabilities, limitations, and qualities of the Machine Learning model [58]. More specifically, it uses OpenAI's commercial web API to access the GPT-3 model [20] to co-create short stories and the titles based on the human user's initial prompt. *Multiverse* responds according to the kind of prompts that is provided by the human users, so to probe AI we wrote textual prompts that embodied concepts and elements from the literary creative writing. This allowed us to intentionally co-create stories with different basic and more advanced elements of the story, such as setting, character, dialogue, metaphors and dialects [46, 70], as well as taking into account the social character of writing as a practice to probe collaborative qualities of AI.

We conducted two progressive autoethnographic studies to respond to our research questions: (1) an autoethnography of co-writing with AI initiated by basic textual prompts using basic elements of fiction such as character, setting, and dialogues; (2) a second autoethnography of co-writing with AI, which builds upon the previous study but employed more advanced textual prompts as elements of narrative including dialects and metaphors undertaken by one of the author of this paper who is trained in creative writing and literature [9]. The autoethnographic research allowed us to synthesize the autobiographical aspects of our personal experiences of crafting a story as a self-reflexive process [37] and the particular experience of human-AI co-writing, with the ethnography of the other – *Multiverse* [60].

[1]https://github.com/motform/multiverse

[2]Two of the co-authors designed and conducted two different autoethnographic studies individually. To communicate the collaborative nature of this project, we decided to use first-person plural "we" for both studies.

Autoethnography is a first-person [13, 51] qualitative research approach, which has emerged as a response to what was viewed as an overly objectivist approach in social sciences in the 1980s. This method provides visibility of ethnographers, allowing them to express their own reflexive first-person perspective regarding a situation, by situating the "ethnographer as the protagonist of the ethnographic narration" [62]. A similar move happened in HCI research, where a humanistic and aesthetic lens was suggested to evaluate subjective experiences of people interacting with computing artifacts [12]. The HCI and design community has been increasingly using the autoethnographic methods as an approach that allows to reflexively evaluate the product of our own design research practice [55]. Autoethnographic research differs from more traditional qualitative enquiries in that it considers the act of doing research as a political and socially-conscious activity [27].

4.1 Autoethnography of co-writing with AI: study #1

As a part of our first autoethnographic study, we as design researchers probed the new version of *Multiverse* through use and co-creating nine stories. We used basic textual prompts as story openings using basic elements of fiction such as character, setting and dialogues (Table 1) — three stories for each element [70]. The reason we focused on the openings was mainly driven by the fact that *Multiverse* is only able to receive the initial prompts from the human user – *Multiverse* is designed as a prompt-based leaning language model [49]. The quality of interaction with the AI is limited and very much determined by that initial prompt provided by the human user. Although the users then can choose between three completions suggested by *Multiverse*, and shape the story line in some ways, the initial opening and framing are the only parts of the narrative that is entirely delegated to the user.

We attended to human-AI cocreation as a dynamic process, by reflecting on questions such as "were the completions suggested by *Multiverse* predictable for the human user (and by particular extension, us as design researchers)?", "did the completions suggested by it help us to be more creative, as did it open the possibilities and creative space?", "what did we learn about ourselves as design researchers and writers through this co-creative entanglement with AI?". We sought to attend to the peculiarities of co-created stories by asking questions such as "did the story include fundamental elements of a narrative, such as a conflict?", "which literary elements were missing and why?", "what does that say about the ways we as design researchers and writers experience a literary work?", "how can we self-reflexively attend to the stories considering the technological determinism in creating them?"

4.2 Autoethnography of co-writing with AI: study #2

In the second autoethnographic study we progressively built upon the previous study results. We probed into the capacities and limitations of *Multiverse* as a creativity tool for creative writing through a series of attempts to co-write stories by a literary scholar—and co-author of this paper. The study was designed in nine parts, each featuring a prompt that was crafted specifically to probe the AI's ability to build from that type of literary element. Literary elements included the introduction of a literary character, a literary setting, a dialogue, a sustained metaphor, etc. (Table 2). As a US-based native English speaker, he included in his initial prompts some specific dialects and American idioms as well. All of these prompts were crafted to probe *Multiverse's* capacity to work with, and to carry forward, that particular element.

After we had briefed the literary scholar in our prior experiences, he then developed the nine prompts, as described above and as shown in Table 2. The *Multiverse* part of the study was conducted in two rounds of two hours (sum of 4 hours). We started the creative writing process, by writing the story openings paragraphs in a separate document, forming the ideas, editing the text several times to craft them intentionally in a way as if we would be writing a story by ourselves and without *Multiverse*. We observed the whole process, reflected during the process about our experiences and feelings. In this second study we were also interested to probe into the capabilities and limitations of the research product as a creativity tool specifically for a competent creative writer. We explored questions like: "Is AI able to pick up the meaning even if the language is ambiguous (Table 2, prompt 3b)?", "Can we sustain or even develop the metaphorical meaning in a story based on a metaphor? (Table 2, prompt 7b)?", "Is it possible for the AI to help maintain a literary image, in a story that features such an image? (Table 2, prompt 8b)?".

We video-recorded the sessions, took notes during and transcribed the sessions after the inquiries. We applied coding, using the thematic categorization method to analyze our data and to reflect on our findings [4].

5 RESULTS

In this section we report the results of our two ethnographic studies, on co-creating stories with *Multiverse* (Tables 1 and 2). We report the findings by reflecting on both the quality of human-AI conversation and the outcome, that is, the co-created short stories. While for the first study we focused more on the general and basic understanding of the experience of co-writing with AI, using different kinds of story openings, in the second study we deployed more advanced literary elements to test the ways AI was able to respond to them.

5.1 The Limits of Not Understanding

We knew heading into the project that GPT-3, like other language models, lacks semantic understanding of the texts it reads and produces [26]. With consumer technologies such as voice assistants — Alexa, Siri, etc. — this limitation is often difficult to discern; it seems as though the system really does understand, even when it does not. But writing fiction is harder than asking Alexa to play a favorite song, or asking Siri for the weather forecast, and so one of our earliest lines of inquiry was to understand how this lack of semantic understanding would manifest in writing fiction. In both studies, we tried openings that took recognizable fictional formats, for example, a character-based opening (Table 1: prompt 2a, 3a), or a metaphor based one (Table 2: prompt 7b).

We observed that often the first or two sentences were closely related to the element; that is, if we started with a setting, the AI's early completions tended to connect to the setting. Afterwards, it would drift away. For instance, in "The Journalist" story, whose

Table 1: Partial list of short stories co-created with GPT-3 and their initial prompts and completions (Study 1)

Prompt	Title	Opening	The beginning of the story (the *italic* text is the initial prompt).
01a	The Blue Knife	Character	*It was sharp and blue.* I looked away from the fish and into the eyes of the man who was selling them. “How much for that one?” I asked, pointing to the fish with the sharp, blue scales.
02a	The Flower	Character	*There was a flower.* It was a beautiful flower, and it grew in the most unlikely of places. The flower was so beautiful that it caught the eye of everyone who passed by.
03a	The Journalist	Character	*Francesca was a journalist, living in Genova.* She was always looking for the next big story to write about. She had been hearing rumors about a big story happening in Milan, and she was determined to find out more.
04a	Amsterdam: The City of Contrasts	Setting	*Amsterdam is a vibrant and colorful city.* With its bicycles, canals and narrow streets, it is a delight to explore. The buildings are a mix of old and new, with gabled roofs and tall spires.
07a	Peter’s Balloon	Dialogue	*May I take that balloon?* Peter asked. The cashier looked at Peter and then at the balloon. “I don’t think so,” she said.
08a	A Night Out	Dialogue	*What do you think about watching a movie tonight?* Azar asked. I don’t think it’s a good idea, Azar, I’m really tired. Azar didn’t seem to [be] happy with my answer.

prompt was character-based, the AI did not develop on the character, her look, her feeling, but quickly moved to the setting and plot (Table 1). Another example was when AI was not able to pick up on the metaphors (Table 2: prompt 7b). In the story “The Writer’s Swim”, we observed that although AI’s first round suggested completions seemed poetic and in-line with the metaphor: “I cannot help but be pulled along by the current, though”. This completion struck our ear as somewhat poetic, but we were aware that the AI didn’t understand or intend poetry; it was just being literal. Our suspicions were confirmed when the next round of completions was not in agreement with the metaphorical meaning of the story. In a third example, the AI was not able to sustain an image (Table 2: prompt 8b). The image of a time-lapse growth of a flower that we introduced in “The Layers of Purple” is understandable for any human reader, but AI not only had difficulty to build on that image, but it introduced some conceptual confusion by suggesting that the flower was not living: “it’s almost like as if it’s alive”.

5.2 Unexpected Competencies

Even as we saw how the GPT-3’s lack of semantic understanding limited its ability to sustain, let alone develop, certain types of literary elements, we did find that it had some pleasant surprises for us as well. One was that it was able to add and introduce secondary characters to the story. That was more salient in the stories that had dialogue (e.g., Table 1: prompt 7a and 8a) and setting as literary elements. For example, in the story “Picking Your Own Chocolate”, the saleswoman and the friend were added to the story by AI. Similarly, in “Peter’s Balloon”, the cashier was a secondary character in conversation with Peter, who was added later. That confirms further that prompt-based models are better capable of being reactive and being engaged in a question-answer kind of interaction rather than a creative conversation.

The AI’s capability to introduce a new character was sometimes highly generative (Table 2: prompt 4b). Especially in the story “A Single Block of Street”, it introduced a new character, “the redhead,” as a guy in the bar with whom Rick (our primary character) fell in love: “One of the guys, a tall, lanky redhead, catches Rick’s eye and motions for him to come over.” Further, the AI was able to introduce a new environmental context to the story that we did not include in the prompt: Redhead’s apartment (prompt 4b). In another story “The Unspoken Conversation” (prompt 6b) we saw that the character developed and defined through the phatic relationships implicit in the AI-generated conversations: “I don’t want a stepfather” or “I’ll try to like him.” These developments changed our own view of the character (in particular, our sense of the boy’s age shifted based on his statements), which influenced our own subsequent decision making. In each case, it seemed that when the AI introduced a new character, it not only did so appropriately, but it was also highly generative for future completions—a crucial quality, given the AI’s tendency to descend rapidly into repetition or absurdity (a topic that we’ll return to later).

A second competency of the AI was its ability to switch appropriately among storytelling modes. One of the most common structural elements in narrative fiction is an alternation between scenes and summaries. Scenes report an event as it unfolds, so for example they include direct quotes, real-time emotional responses,

Table 2: List of short stories co-created with GPT-3 and their initial prompts and completions (Study 2)

Prompt	Title	Opening	The beginning of the story (the *italic* text is the initial prompt).
03b	The Sun Will Rise	Setting (Ambiguous condition)	*It was the sunlight that would finally wake him. As he rolled over, he felt wetness.* He looked over and saw that his wife was crying. He asked her what was wrong, but she just shook her head and turned away. He rolled back over and tried to go back to sleep, but the sunlight was too bright.
04b	A Single Block of a Street	Setting (Milieu)	*Hungtington's gay neighborhood is not so much a neighborhood as it is a street, more precisely, a single block of a street, near but not quite in the downtown area. It features a single bar, though the only sign of its prevailing identity is a faded, wallet-sized rainbow flag decal in one of the arched windows. Stepping inside its trendy entrance, sounds of music and laughter drown out the shouted orders at the bar. A generation ago, smoke would have hung in the air, making slow dances around the neon lighting, but smoking is now prohibited. Lit now by the harsh red and yellow, Rick pours a bourbon into a small glass with ice.* As he takes a sip, he watches the people on the dance floor and wonders who is in love and who is just pretending. A group of guys walks in and takes the space on the dance floor in front of Rick.
05b	Sugar-High Sundays	Setting (Repetitive events)	*Every Sunday, before we would go to Mass, Mom would go to the bakery and buy a dozen doughnuts, bringing home a box of jelly-filled sugar, Boston crème, and maple glazed doughnuts. By the time she got home and opened the box, powdered sugar, chocolate, jelly, and cream would be liberally smeared on the doughnuts and box alike, but we children wouldn't care, as we fought over who got what. Later, powered by a collective sugar-high, we would pile into our red station wagon, with wood paneling on the sides, and drive to St. James Church, where, if we were lucky, Father Martin rather than Father John would offer Mass. We would all be so excited to go to mass, especially because we knew that Father Martin would offer a longer mass.* We would all be so excited to go to mass, especially because we knew that Father Martin would offer a longer mass. This would give us plenty of time to eat our doughnuts.
06b	The Unspoken Conversation	Setting (An event)	*After he left, I replayed the evening: his arrival, the wine, the dinner, and our conversation. I had wanted to talk to him about Freddie. Before we take the next step, I needed to know that he and my son will get along. More than that: I wanted them to be like father and son. Yet every time I tried to bring up the topic, Todd was vaguely reassuring and somehow the conversation drifted away, without a real conclusion.* I was getting frustrated; I wanted to know what Todd's plans were for Freddie. I decided that I needed to talk to Freddie about Todd.
07b	The Writer's Swim	Theme (Sustaining a metaphor)	*Writing is like swimming. I am immersed in something greater, more forceful than me; something that came before and will survive long after I do. Yet I am not powerless; whether language or waves, I can push it, shape it, move through it. As I do, perhaps I get somewhere; perhaps I just get a feel for what it's like to be over here.* I cannot help but be pulled along by the current, though. It's a form of exploration, a way to understand the world and myself a little better.
08b	Layers of Purple	Theme (Sustaining an image)	*The flowers burst above the ridged glass vase. Each blossom's five purple petals offer a star-like contrast to the leafy green steams beneath them. The blossoms number in the dozens, perhaps a hundred. Their impact comes their patterns and repetitions, as layers upon layers of the purple flowers push each other aside, each on an imperceptibly slow, yet urgent and occasionally violent, journey towards the light.* I can't help but be mesmerized by their beauty. I can't help but be mesmerized by their beauty, even though I know they won't be around for long. I watch as a bee lazily circles one of the flowers, before landing on its petal. I'm not the only one who is mesmerized by their beauty.

and so on. Summaries compress time and sketch out key information that happens between scenes. We found that the AI could make these shifts on its own. For example, such a shift happened in "The Unspoken Conversation" story (prompt 6b). The story opened as a conventional summary of an event that happened once, and the intention was to see whether AI can pick up on that literary element: "After he left, I replayed the evening: his arrival, the wine, the dinner, and our conversation." We observed that AI can actually pick up on the literary element, but we also saw that the fourth round of completions introduced a dialogue, which not only helped to continue the story but also displayed a shift from a summary to a scene: "'Freddie, I need to talk to you about Todd', I said. 'What about him?', Freddie said". As with the AI's introduction of the redhead in "A Single Block of Street," this narrative contribution was both appropriate to the story and generative, because the AI was able to continue the conversation for several additional completions.

The shift from summary to scene also entailed another AI contribution, in which it showed (apparent, if not real) understand of the story's underlying concepts. The story prompt indirectly suggests that the narrator is intending to marry Todd, and she is worried about how Todd and her son, Freddie will interact. AI completions introduce the concept of a "stepfather," which is correct and appropriate, but which was never made explicit in the prompt. The AI also was able produce completions that developed this *conflict* in an appropriate way, having Freddie say, "I told him that he did not have to call Todd his stepfather [...]".

Although we know that the AI does not form a semantic understanding of the language it processes, and while evidence was abundant that demonstrated this fact, nonetheless, the AI was on many occasions able to make appropriate literary moves, and when it did, it was also generative in the sense that it could produce relatively high quality completions for the near future (e.g., the next 3 or so completions) when it did make such moves.

5.3 (Un)Predictability

Predictability is a double-edged sword in the context of a system designed to support creativity. On the one hand, a certain amount of predictability is desirable, because it helps the user to build competence with and confidence in the system. On the other hand, creativity is often stimulated by surprises—new ways of seeing a familiar thing, or new juxtapositions of ideas, the possibility of material "back talk" [63], and so on. In conventional creativity support systems, such as image editing software, products have become predictable as software systems (e.g., with consistent UIs, keyboard shortcuts, and actions), while still providing room for experiments and surprises with a given image itself.

With a system built around human-AI collaboration, the "smart" nature of AI responses adds a new dimension of (un)predictability. One of the ways that we probed this issue was to write prompts that were deliberately ambiguous. In "The Blue Knife," for example, the prompt was "It was sharp and blue," without specifying what exactly it was. In the AI-created title, it resolved the "sharp and blue" object as a blue knife. But in the narrative completions, it resolved it as a fish: "I looked away from the fish and into the eyes of the man who was selling them." Taken individually, both inferences are reasonable: knives and fish are sharp. Yet the AI surprisingly put them together: the sharp and blue object is both a knife and a fish. A human reader—or a human collaborator—can of course dismiss this as absurd. Another, more intriguing, possibility, however, is to imagine that the AI was proposing a metaphor: the fish as knife. In such a scenario, perhaps someone later in the story will get cut on the fish, or the fish's sharpness might even be used as a weapon. Again, the AI doesn't understand what it is producing, so it is easy to view moves like this as absurd (and doing so is not incorrect). But in a creative situation, as literary composition is, such absurdities can themselves become generative.

A second story with an intentionally open-ended prompt was "The Sun Will Rise," which has its protagonist "finally" woken by the sun in a wet bed. There can be many explanations for a wet bed—sweat, urine, a leaky roof—but in this case the AI resolved it to wetness from the tears of a second character that it introduced, the protagonist's wife. As soon as it had done so, it then began to generate a story about those tears—that the wife was sad, that the husband wanted to comfort her, and so forth. In short, the AI's unpredictability was generative not only for human users, but even for itself. The converse was also true: predictability became un-generative. That is, we found that when AI completions were highly predictable, the narrative was in danger of becoming stuck. What we mean by that is that its completions would endlessly repeat themselves, sometimes verbatim, sometimes in very close paraphrases, to the point that it offered no completions that offered anything other than repetition. Indeed, such a repetitious loop is eventually what ended "The Unspoken Conversation" (Table 2).

5.4 Absurd Semantic Connections and Structures

Based on our experience, which was in-line with [26, 28], the AI often struggled or failed to achieve appropriate semantic relationships and misunderstood semantic concepts. Sometimes it created new and unrelated concepts, such as the knife-fish described earlier; other times, it created other inappropriate connections.

One example occurred in the story "High-Sugar Sundays" (prompt 5b). In it, the narrator describes how on Sundays his family would enjoy doughnuts before heading to Mass; in the narrative, breakfast and Mass were clearly two different moments in time. But the AI confused the separate temporal moments: "We would all be so excited to go to mass, especially because we knew that Father Martin would offer a longer mass. This would give us plenty of time to eat our doughnuts." The AI completions made the surprising assertion that children preferred longer Masses—which, frankly, seems less probable than the opposite—because that would give them more time to eat doughnuts, thereby moving the breakfast into the Mass itself.

A similar moment of hilarity happened in "The Sun Will Rise" (prompt 3b), which starts: "It was the sunlight that would finally wake him. As he rolled over, he felt wetness." As noted earlier, this prompt was intentionally vague, but the AI chose tears as the cause of the wetness and generated a narrative in which the husband tries to comfort a weeping wife. Yet the AI seemed to want to hang onto the image of the sun, and so it brought the sun back into the narrative, with the following completion: "And so it was that he took her away, to a place where the sun never shines." In American slang, “a place where the sun never shines” is a rude idiom. So, while one can understand what the AI was doing—attempting to connect the semantic concepts of sun and sadness–it instead finished off the story with a hilariously rude ending.

5.5 “Keeping the AI alive”

We have shown in the preceding sections that the AI gets into trouble when it over-uses intensifiers, confuses semantic concepts, and makes inappropriate connections. All of this calls to mind that one of the greatest challenges with the first study was that stories tended to end quickly, because the AI became repetitive or dipped into absurdity without any possibility of recovery. One of the goals of the second study was to try to experimentally discover the conditions by which the AI could produce viable narrative, so that that process could be prolonged sufficiently to actually tell a story that doesn’t short-circuit in 2-3 sentences.

As we began to discover what some of those conditions were, we began to make real-time decisions to maximize them, a process we colloquially referred to as “keeping the AI alive.” Because getting beyond 2 or 3 completions was so fundamental to any possibility of success, keeping the AI alive began to override any literary goals we had.

Specifically, we noticed that the majority of AI completions were of one of the following two types: (1) *The introduction of something new to the story: the introduction of a new character, a new concept or action;* (2) *The development of something existing, or what we called “intensifiers”: a description of how someone felt about something, or reacted to something, or a description of why something was important or what it meant.* While hardly universal, in general, choosing a completion from the first category is more likely to keep the AI alive than choosing a completion from the second category. We surmised that the reason is that completion in the first category explicitly added some new semantic content—especially new words—which gives the AI more options to work with, while completions from the second category, which *develop* something in the narrative, while often of great importance to character and thematic development in a narrative, led to dead-ends because they did not add new words.

This led to dilemmas, where the system would offer completion from each category, in which the immediate literary goal would be better served with an intensifier, while the keeping the AI alive goal would be better served with the new line of plot development. Faced with one such dilemma, we chose the completion that advanced the plot, and we were rewarded when in the next round of completions in “The Unspoken Dialogue,” the AI introduced a line of dialogue: "Freddie," I said, "I need to talk to you about Todd." We thereby experienced a kind of improvement in our interactions with *Multiverse* after a while. That does not mean that we created better stories based on our literary judgments, but rather that our story lengths did increase without excessive repetition or absurdity. Of course, that implies that we de-centered our own judgments and accommodate the peculiar needs of the AI. We even began to experience accommodating the peculiar needs of the AI as though it were a care relationship. Indeed, at one point in the conversation we compared writing fiction with the AI to taking care of a Tamagotchi. This analogy—writing fiction as Tamagotchi care—is one of the more surprising outcomes of both studies.

6 DISCUSSION

6.1 Literature as a “Practice”

It is well known that AI language models (e.g., GPT-3) do not build semantic understandings of texts, and this is often used to explain some of the problems it has in producing fictional discourse—problems that we experienced in abundance. But there is another, less remarked, problem, which might be even more fundamental. Writing fiction is a *practice* [45]; that is, writing fiction is a conventions-governed activity that happens in the social world, by real actors, which includes above all authors and readers (but also critics, booksellers, etc.). The AI is not a part of, nor does it understand, this social world. Philosophers of art Lamarque and Olsen argue that what makes a work “fiction” is not about whether it is true (since fictions are filled with truths, and allegedly truthful discourses are filled with falsehoods); rather, what defines fiction is storytellers’ and audiences’ participation in a social practice, which is rule-governed both in terms of the conventions of narrative discourse and in the attitudes needed to engage them appropriately ([45] p.32 and 33). Common examples of such conventions include the following: fictions are grouped together in bookstores, their covers’ graphics, plot summaries and critic reviews mark them as fiction, and by the text itself (e.g., “In a hole in the ground there lived a hobbit”). GPT-3’s lack of participation in this world is much more fundamental and consequential than its lack of ability to form semantic understandings. At issue is not only that GPT-3 doesn’t understand what a word means; what is that issue is that it cannot engage in social acts such as the bedtime story, the reading group, or a religious reading; nor can it perform what Coleridge famously called “the willing suspension of disbelief”; AI cannot understand that we create fictions to explore and critique aspects of our social worlds; it cannot take up the “fictive stance” (the capacity to make-believe, imagine, pretend); it cannot judge a given work’s place among the canons of literary fiction.

And yet, GPT-3 unmistakably can produce text that appears to reflect certain competencies of fictional practice. We saw this in elements of both the story and its telling. For the former: GPT-3 introduced characters, settings, situations, actions, conflicts, with appropriateness a fair amount of the time. For the latter: it could switch between scenes and summaries appropriately; it picked up on unique tenses (e.g., the past imperfect); it was able to sustain point of view (e.g., first-person). Our inference is that social conventions and purposes often manifest themselves with distinctive textual features. In fact, after the project ended, just to test this inference, we identified a highly conventional story opening, and we put it to *Multiverse:* “Once upon a time.” This opening is conventional to fairy tales, which are often simple children’s stories that take place in a medieval like setting (e.g., the setting of Disney’ *Snow White*). *Multiverse* wasted no time, naming the story (before even a single completion had been selected), “A Tale of Two Kingdoms,” and the first completion offered was, “A Princess fell in love with a peasant boy.”

In short, while the AI is oblivious to the social world of fairy tales, it can positivistically pick up on explicit textual features that are statistically common in texts that are a part of that social world. This, in turn, implies that the more explicitly conventional the prompt is, the more likely the AI language model will be able to reproduce those conventions competently. That is useful if the literary goal is "keeping the AI alive," but hewing to literary cliches is also not a recipe for the inventiveness that we associate with good fiction.

6.2 Human-AI Relationships in Co-writing Fiction

Characteristics of creativity, such as agency, supports, and artifacts that were mentioned earlier [8] are helpful to understand the changes and developments in the creativity tools and the dynamics of co-creative activity with AI. We observed for example that agency was distributed in the act of initiation of the creative activity (users' prompts), the creation process and the continuous user's decision making along the process. However, unlike more traditional creativity support tools, the AI had a more prominent role in creating and shaping the result—we found ourselves being in the front of a series of limited options to choose from, sometimes unrelated, sometimes repetitive and sometimes even absurd, so we felt restricted to comply with AI's higher-level decisions [34]. That led to a form of de-construction and re-construction of the creative process in which the human user was de-centered, in that the goal shifted from telling a good story to keeping the AI alive in the process!

Putting AI at the center of the process—not by choice, but by imposition—led to a series of insights regarding human-AI relationships in creative writing. Superficially, it seemed to invert the relationship between technology and the human, with the human user serving the technology. But the relationship was more complicated than that. As in the master-apprentice model, in which the master guides or helps the apprentice to learn a practice, we felt that we, like masters, were "schooling" the AI to write; we never expected good literary fiction to emerge, but we wanted to assess where the AI was at this point in time, and then we wanted to contribute to AI research in this area. Second, we self-reflexively viewed that one of our most powerful drives to pursue a creative collaboration with AI on writing, was a kind of curiosity towards the Other, this non-human actor that is often competent at producing fictional discourse following logics and achieving results—good and bad—that exceed our understanding. And as that Other was also a collaborator, this relationship surprisingly evolved into a relationship of care. Finally, we reflexively considered the politics of our own sentience and intelligence as design researchers collaborating with an AI. For if this was a human-AI collaboration, it was nonetheless one that featured asymmetric power: we set the prompt, we determined when the AI was doing well or poorly, and we analyzed the AI's capabilities and limitations. Reframing the collaboration as "keeping the AI alive" was, among other things, a tacit assertion of our intellectual superiority.

6.3 Synthesis of Disciplinary Goals

During the course of the study, we gradually realized that, with regard to AI, we needed to distinguish between (1) technology goals and (2) literary goals. Technology goals include research that advances the development of language models so that the AI is able to summarize and translate human texts, answer human questions, and generate text for human use; as well as to develop effective writing tools and positive user experiences. Literary goals, on the other hand, can be quite different: to support fine expression; to contribute to and/or develop new literary subgenres, including styles, conventions; to help produce literary components, such as characters, plot lines, images, themes, settings. That means, for example, that what might be a problem for data science may not at all be a problem from a literary standpoint. For example, many modernist artists used cut-up text and other random methods to support creativity; thus, a given absurdity from AI might be viewed as a failure from the perspective of a data scientist and a success from the point of view of a literary artist [3].

We also came to grasp that technology and literary goals started to synthesize into a third one, that of exploration of limits and capabilities of *Multiverse* driven by a curiosity towards the Other. We call that third goal a design research goal: an exploratory, iterative, and constructive goal set to not only embrace the pragmatics of the situation, the constraints, and capabilities of the technology, but also to pursue a better user experience. Here, we found ourselves in the front of many barriers, including technological barriers to achieve a "good" quality literary work as the outcome of our creative collaboration with AI. That felt frustrating, sometimes even disappointing, but we were driven by a purpose that seemed to go beyond of creating a fine creative piece, but muddling through the contingencies of the circumstance, synthesizing goals and materials, crafting what is possible through a reflective conversation with the situation [32, 63].

7 CONCLUSION

Natural Language models, like GPT-3, are increasingly part of our everyday lives and our creative activities. We conducted two progressive autoethnographic studies to self-reflexively and creatively engage *Multiverse,* a research product that uses GPT-3 model to create a collaborative space to co-create stories with human users. We found that our relationships with AI were driven as much by curiosity to learn about our mysterious collaborator, as they were by any hope to use AI as a tool for creativity. We also observed that we built a relation of care with AI, similar to that with Tamagotchi. Further we moved past the question of whether AI could write a good piece of fiction to consider with much more granularity what the AI could do. We found that the AI was capable of producing appropriate, novel, and even interesting fictional elements, both elements inside the world of the story (e.g., characters and actions) and elements of the telling of the story (e.g., the alternation between scene and summary). We believe that there is much more potential in such design-driven and humanistic perspectives in Human-AI co-creation and that further inquiry in this mode might hold powerful insights into still uncharted areas of human-AI literary collaborations.

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